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Bright light and mental fatigue: Effects on alertness, vitality, performance and physiological arousal

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ABSTRACT

Alertness-enhancing effects of bright light are particularly strong at night or after sleep deprivation. Alerting effects during daytime also exist, yet these appear to be more modest. In this study, we investigate whether a higher illuminance level particularly benefits individuals who suffer from mental fatigue – not from sleep pressure, but from mental exertion. A 2 × 2 within-subjects design ($N = 28$; 106 sessions) was applied to investigate effects of 1000 vs. 200 lx at the eye on self-report measures, task performance and physiological arousal after a mental antecedent condition (fatigue vs. control). Results showed that participants felt less sleepy, more vital and happier when exposed to bright light. Effects on subjective sleepiness and self-control capacity were stronger under mental fatigue. Vigilance benefited from bright light exposure – although this effect emerged with a delay irrespective of the antecedent condition. Other tasks showed more mixed and sometimes even adverse effects of bright light.
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1. Introduction

As humans evolved as a diurnal species, their psychological functioning is primarily less optimal in the late evening or at night compared to daytime hours (e.g., Cajochen, Kräuchi, & Wirz-Justice, 2003; Dijk, Duffy, & Czeisler, 1992; Hull, Wright, & Czeisler, 2003; Van Dongen & Dinges, 2005). However, persons may also experience fatigue and resource depletion during daytime, even in the absence of sleep deprivation. During the day, we use and deplete mental resources (Hartig & Staats, 2003; Kaplan & Kaplan, 1989). Engagement in many work-related and demanding tasks requires focused attention, motivation and allocation of mental resources (Baumeister, Vohs, & Tice, 2007; Inzlicht & Schmeichel, 2012; Kaplan & Berman, 2010; Meijman & Mulder, 1998). Accumulation of effort spent throughout the workday might result in increased feelings of sleepiness, lack of energy, loss of self-control, psychological stress and decrements in motivation and task performance (e.g., Baumeister et al., 2007; Hagger, Wood, Stiff, & Chatzisarantis, 2010; Inzlicht & Schmeichel, 2012; Kaplan & Berman, 2010; Meijman & Mulder, 1998).

Research has shown that the physical environment may affect the level of mental fatigue people encounter. Several studies have, for example, demonstrated that context and ambient conditions, such as office layout, indoor plants or views to the outside, can revitalize office workers and help them recuperate from stress or mental fatigue (e.g., De Kort, Meijnders, Sponselee, & IJsselsteijn, 2006; Dijkstra, Pieterse, & Pruyn, 2006; Hartig, Böök, Garvill, Olsson & Garling, 1996; Ulrich et al., 1991; Veitch, 2011). This suggests that environmental features of an office environment have the potential to influence individuals' experienced level of mental fatigue, vitality level and performance. Several field studies have shown that lighting too is a potential environmental feature impacting office workers' mental wellbeing, behavior and performance. More specifically, these studies indicated that the color of white lighting and the type of light source can affect employees' feelings of alertness, self-reported performance and their need for recovery during regular working hours (Mills, Tomkins, & Schlangen, 2007; Smolders, De Kort, Tenner, & Kaiser, 2012; Viola, James, Schlangen, & Dijk, 2008).

In addition, a large body of research has demonstrated acute activating effects of bright light exposure on subjective and objective indicators of alertness and arousal. Most of these studies investigated effects of light intensity under conditions of relatively high fatigue and sleep pressure. Several laboratory studies have, for example, demonstrated that exposure to higher illuminance levels at night resulted in lower levels of melatonin secretion, increased physiological arousal, higher subjective alertness, and improved

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sustained attention and cognitive task performance (e.g., Badia, Myers, Boecker, & Culpepper, 1991; Boyce, Beckstead, Eklund, Strobel, & Rea, 1997; Cajochen, Zeitzer, Czeisler, & Dijk, 2000; Campbell & Dawson, 1990; McIntyre, Norman, Burrows, & Armstrong, 1989; Myers & Badia, 1993; Rüger, Gordijn, Beersma, de Vries, & Daan, 2006; Rüger, Gordijn, de Vries, & Beersma, 2005). Similar beneficial effects of bright light exposure during daytime have been demonstrated for individuals who had first experienced substantial light and/or sleep deprivation (Phipps-Nelson, Redman, Dijk, & Rajaratman, 2003; Rüger et al., 2006; Vandewalle et al., 2006). Together, these studies indicate that light can benefit persons under conditions of high circadian and/or homeostatic need for sleep and when they are particularly sensitive to light due to prior exposure to very low light levels.

A few recent studies also showed alerting effects of bright light on healthy day-active persons during regular working hours. A field study by Smolders, De Kort, and Van den Berg (2013) provided support for a direct link between exposure to more intense light and feelings of vitality during daytime and in everyday situations. Their results showed that hourly light exposure was a significant predictor for feelings of vitality, indicating that persons who were exposed to more light experienced higher feelings of vitality, over and above the variance explained by person characteristics, time of day and activity patterns. In addition, a recent laboratory study showed that even in the absence of sleep and light-deprivation, exposure to a higher illuminance at eye level can induce subjective alertness and vitality, increase physiological arousal and improve performance on a sustained attention task (Smolders, De Kort, & Cluitmans, 2012). Together, these studies suggest that – although the effects were more modest than at night – exposure to a higher illuminance level may also (temporarily) help overcome tiredness and decreased vitality during regular daytime hours. In fact, the field study by Smolders et al., 2013 showed that the relationship between light exposure and vitality was most pronounced when participants experienced relatively low vitality during the previous hour.

In the study by Smolders, De Kort, and Cluitmans (2012), effects on self-report measures of sleepiness and vitality and on heart rate were immediate and consistent during the hour of bright light exposure. In contrast, the effects of a higher illuminance level on task performance and heart rate variability were dependent on duration of exposure: These effects were most pronounced towards the end of the 1-h long phase of testing and light exposure. A potential explanation for the delayed effect of bright light on performance found in the laboratory is that more intense light improves performance mainly when persons suffer from sleepiness and resource depletion. This is consistent with research showing that light exposure at night or among sleep-deprived participants can improve reaction times immediately (Lockley et al., 2006; Phipps-Nelson et al., 2003). Indirect indications for this also come from a study performed by Vandewalle et al. (2006), who investigated daytime effects of bright light on task performance and brain activity using fMRI. These authors excluded participants who did not show a response in alertness to exposure to a higher illuminance. In the supplemental analyses they reported that these participants already had faster response times at baseline than participants who did respond to light, suggesting that they already were very alert. An alternative explanation for the delayed emergence of bright light effects on performance and heart rate variability could be that effects on these variables need extended exposure to bright light, due to required exposure thresholds or (relatively) slow activation mechanisms (e.g. Vandewalle, Maquet, & Dijk, 2009).

In the current laboratory study, we test whether bright light particularly benefits individuals' level of alertness and vitality during daytime when they suffer from fatigue due to mental exertion. We thus compared effects of an illuminance manipulation (200 lx vs. 1000 lx at the eye) during daytime between situations in which persons' resources had been mentally depleted and in which their resources were not depleted prior to the light exposure. As earlier studies have suggested that effects of bright light may differ depending on the type of measure (e.g., Rüger et al., 2006; Smolders, De Kort, & Cluitmans, 2012), we investigated effects of daytime light exposure on subjective sleepiness and vitality as well as objective indicators of task performance and autonomic nervous activity. If the effects of bright light on feelings, performance and physiology are dependent on a person's prior mental state, we should see both stronger and more immediate effects of the illuminance manipulation when participants experience fatigue. In contrast, we should see less pronounced effects for rested participants, i.e., more subtle or delayed effects. Based on earlier findings of light exposure patterns and vitality in real-life situations (Smolders et al., 2013), we expected this interaction for the self-report measures. In addition, based on findings on daytime exposure to a higher illuminance level in the laboratory (Smolders, De Kort, & Cluitmans, 2012), we hypothesized that if mental fatigue indeed was responsible for the delay in bright light effects there, we should also find a significant interaction between illuminance level and individuals' prior mental state on task performance and heart rate variability in the current study. In addition to the potential state-dependent effects of bright light on alertness and arousal, we investigated whether individuals' mood as well as their appraisals of the lighting, experience of the space and beliefs concerning the effect of the lighting on performance and mood differed as a function of the lighting condition and their mental state. The current study will thus provide insights in whether persons' sensitivity to bright light effects on experiences, performance and physiology during daytime depends on their psychological state.

2. Method

2.1. Design

A 2 × 2 within-subjects design was applied to test effects of two illuminance levels (200 vs. 1000 lx at eye level, 4000 K) under two antecedent conditions (Fatigued vs. Control). Participants came to the lab on four visits on separate days during the same timeslot in the morning (9:00am, 10:20am or 11:45am) or in the afternoon (1:15pm, 2:45pm or 4.15pm). Every session started with a 7-min baseline phase and a 29-min mental fatigue vs. control manipulation under the same lighting condition (200 lx and 4000 K at work plane), followed by exposure to one of the experimental lighting conditions for 30 min. The order of conditions was counterbalanced across participants. There was no daylight contribution in the room during this experiment.

2.2. Participants

Twenty-eight students participated in this laboratory study, of which 12 were male and 16 female (mean age 23, $SD = 4.1$, range 19–39). Four participants were not able to participate in the fourth session and in two sessions the lighting did not work properly, resulting in 106 75-min sessions. None of the participants were extreme chronotypes according to the Munich Chronotype questionnaire (MCTQ; Roenneberg, Wirz-Justice, & Merrow, 2003), nor did they have eye complaints or complaints about their general health.

2.3. Procedure

Before the start of the first session, participants signed a consent form. At the start of each session, participants applied electrodes for heart rate and skin conductance measures according to the

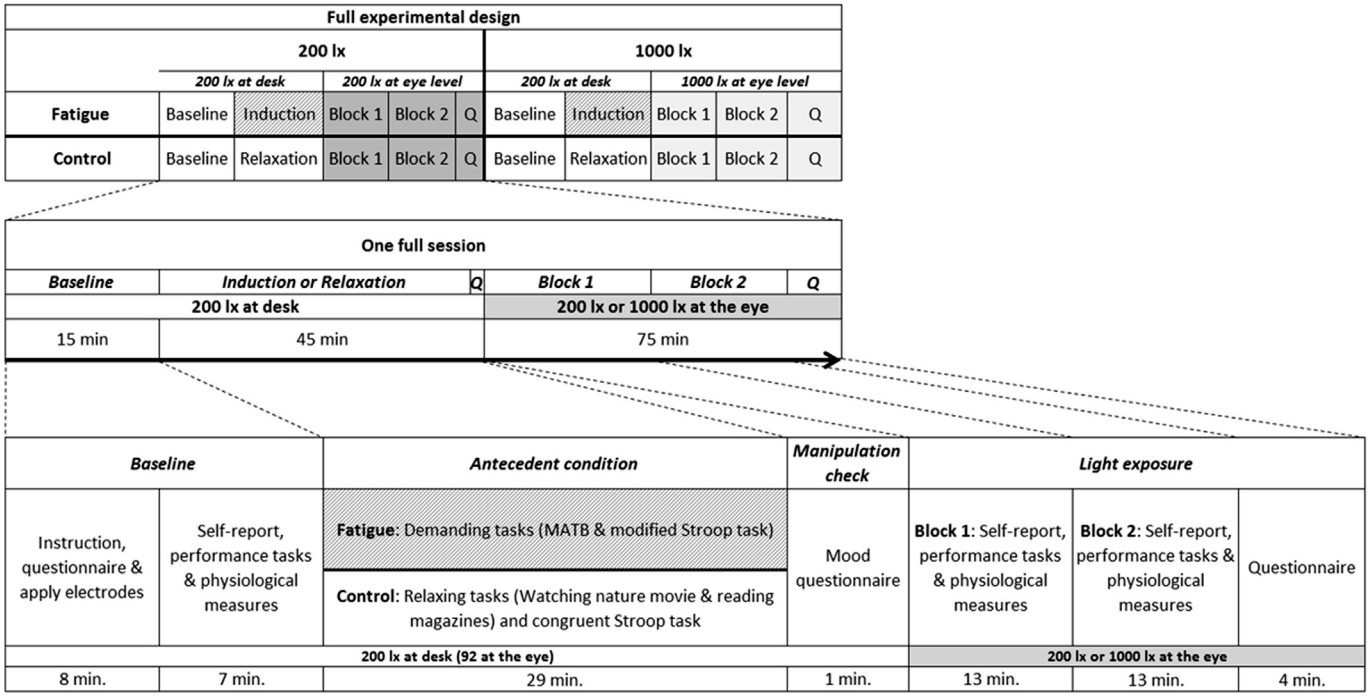

Fig. 1. Overview experimental conditions and procedure of one full session.

instructions given by the experimenter. Subsequently, every session started with a 7-min baseline measurement phase consisting of a 1-min resting period, performance tasks and a short questionnaire (see Fig. 1). Baseline performance was measured using three different tasks: A 3-min auditory Psychomotor Vigilance Task (PVT), a 1-min auditory Go-NoGo task and a 1-min 2-Back task. After the baseline measurements, the mental fatigue vs. control manipulation started, which took 29 min. After this manipulation, participants completed a short questionnaire (manipulation check). During the baseline measurements and fatigue induction, participants always experienced the same baseline light conditions (200 lx at the desk).

After the fatigue vs. control treatment, two additional wall luminaires were turned on and participants were exposed to either 200 lx or 1000 lx (at the eye) for 30 min. During this light exposure, self-report scales, task performance and physiological measures were administered in two repeated 13-min measurement blocks. Each block started with a 1-min rest period. Subsequently, performance was measured with a 5-min auditory PVT, a 3-min auditory Go-NoGo task and a 3-min 2-Back task. At the end of each block, participants completed a short questionnaire to measure sleepiness, vitality, tension, positive affect and negative affect.

At the end of each session, participants completed questions concerning subjective state self-control, their evaluation of the lighting condition and the environment, time of going to sleep the night before, time of awakening and time spent outside. In addition, at the end of the last sessions, questions concerning person characteristics, such as light sensitivity, chronotype and trait vitality, were administered. Every session lasted roughly 75 min and the participants received a compensation of 12.50 Euros per session. The study was approved by the board of the Human Technology Interaction group in Eindhoven and took place from February to June 2012.

2.4. Setting

The laboratory where the experiment took place consisted of two rooms divided by a curtain. The two simulated office environments had a size of 3.2 m by 1.8 m. The room was equipped with surface-mounted Philips Savio luminaires at the walls and recessed Philips Savio luminaires in the ceiling. The wall luminaires (Philips Savio TBS770 6x14W/827/865 HFD AC-MLO CVC) contained six fluorescent tubes of 14 W, of which four tubes of 6500 K (TL5 HE 14W/865 SLV) and two tubes of 2700 K (TL5 HE 14W/827 SLV). Each ceiling luminaire (Philips Savio TBS770 3x54W/827/865 HDF AC-MLO CVC) contained three fluorescent tubes of 54 W, of which two tubes of 6500 K and one tube of 2700 K. All luminaires have an acrylate micro-lens optic cover which blends the two lamp types to create a virtually homogeneous luminous surface. The light intensity, spectral power distribution (SPD) and color rendering index (CRI) were measured with a calibrated spectroradiometer (JETI Specbos 1201) at eye level (vertically, height 1.20 m) aimed at the wall in the gaze direction of the participant. The SPD of the lighting at 4000 K is depicted in Fig. 2. The CRI at 4000 K was $R_a = 86$.

During the baseline, the ceiling luminaires provided basic room illumination (200 lx on the work plane; 92 lx at eye level (photon

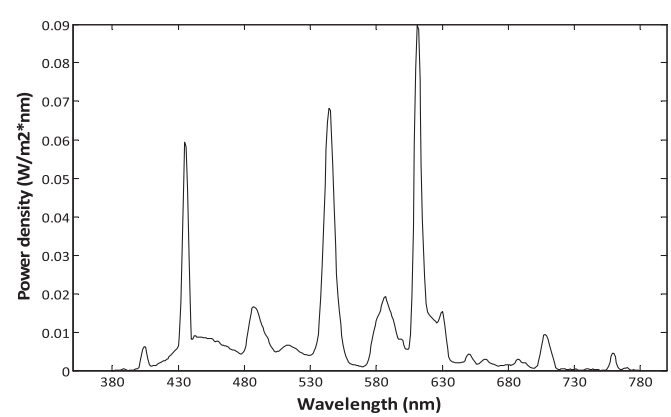

Fig. 2. Spectral power distribution measured at eye level in the 1000 lx (4000 K) condition.

density: 1.61×10^{14} photons*s^{-1}*cm^{-2}; irradiance: 29 μW/cm^2)). Upon the start of the experimental light treatment, two wall-mounted luminaires in front of the participant were turned on and the illuminance of the ceiling lighting was increased. In the 200 lx condition, settings were such that this resulted in a total of 200 lx at the eye (photon density: 1.80×10^{14} photons*s^{-1}*cm^{-2}; irradiance: 68 μW/cm^2) and 280 lx on the work plane. In the 1000 lx condition, this resulted in 1000 lx at the eye (photon density: 8.70×10^{14} photons*s^{-1}*cm^{-2}; irradiance: 330 μW/cm^2) and 1040 lx on the work plane. In both conditions, the CCT was set at 4000 K. The main furnishings of the room consisted of a desk with computer, chair and a cabinet. The walls were off-white and had a reflectance of 88%, the ceiling had a reflectance of 96.2%, the floor was dark gray with a reflectance of 5% and the desk was gray and had a reflectance of 55%. A window (1.6 × 1.6 m, east oriented) was covered with a roller shutter outdoors and a blackout blind indoors, preventing any daylight penetration.

2.5. Mental fatigue induction

Mental fatigue was induced with two demanding tasks: a 9-min Multi-Attribute Task Battery (MATB; Santiago-Espada, Myer, Latorella, & Comstock, 2011) and a 20-min modified Stroop task (Wallace & Baumeister, 2002). The MATB is a demanding multi-task using a flight simulation in which the participant keeps track of multiple parallel processes (maintaining the volume in two fuel tanks, repairing the fuel system when broken, monitoring the aircraft, tracking the aircraft with a joystick, and adjusting the communication channel when needed). Participants were instructed to keep track of all parts and perform the tasks as fast and accurately as possible.

After the MATB, participants engaged in a modified Stroop color-naming task. Words describing a color (red, yellow, green, or blue) were presented one by one in an incongruent-color ink on the screen. Each word was presented for 1 s with a 2.2-s interval between the words. During this interval, the screen remained gray for 1500 ms, then a fixation point appeared for 500 ms, the screen went gray again for 200 ms, and subsequently the new word was presented. For each word, participants named the color of the ink by pressing the corresponding key on the keyboard, except when the word was presented in red (in one version or in yellow in another version). In these cases, participants had to press the color indicated by the word instead of the color of the ink.

In the control condition, participants engaged mainly in restorative tasks, i.e., relaxing tasks requiring relatively low mental effort: they watched a 9-min nature movie and then read leisure-time magazines for 17 min. At the end of the control condition, participants engaged in a 3-min Stroop task with congruent trials, i.e., with words describing the color of the ink it was written in (see Fig. 1).

2.6. Measures

Both self-report and objective measures were employed to assess alerting and vitalizing effects of the light manipulation (illuminance level at the eye) after the mental fatigue induction vs. control condition. In addition, participants' mood as well as their appraisals and beliefs concerning the lighting condition were measured.

2.6.1. Self-reported sleepiness, vitality, mood and self-control

Subjective sleepiness was measured with the Karolinska Sleepiness Scale (KSS; Åkerstedt & Gillberg, 1990). The response options of this scale ranged from (1) 'extremely alert' to (9) 'extremely sleepy – fighting sleep'. Subjective vitality and tension were assessed with six items selected from the Activation-Deactivation checklist (Thayer, 1989). Four items ('energetic', 'lacking in energy' (reversed), 'alert' and 'sleepy' (reversed)) assessed subjective vitality and two items assessed tension ('tension' and 'calmness' (reversed)). The internal reliability of these scales were $\alpha = .90$ and $\alpha = .59$, respectively. Positive affect ('happy') and negative affect ('sad') were each measured with a single item. The response scales of these eight items ranged from (1) 'definitely not feel' to (5) 'definitely feel'.

Subjective state self-control was assessed at the end of each session using six items selected from the State Self-Control Capacity Scale (Ciarocco, Twenge, Muraven, & Tice, 2011). Participants indicated whether they were (a) mentally exhausted, (b) would give up easily on a difficult task, (c) felt drained, (d) experienced low willpower, (e) had difficulties controlling urges, and (f) whether it would take a lot of effort to concentrate, on a 7-point response scale ranging from (1) not true to (7) very true. This scale was internally consistent with $\alpha = .92$.

2.6.2. Task performance

Three tasks were employed to assess different aspects of performance. A 5-min auditory Psychomotor Vigilance Task (PVT) assessed sustained attention (Loh, Lamond, Dorrian, Roach, & Dawson, 2004). During this test, beeps of 400 Hz were presented at random intervals of 1–9 s and the participant had to press the spacebar as fast as possible after hearing a beep. An auditory Go-NoGo task measured inhibitory capacity. In this task, beeps of 400 Hz and 600 Hz were presented at random intervals of 1–9 s and participants pressed the spacebar as fast as possible only after hearing a 400 Hz beep, and not after hearing a 600 Hz beep (20% of the cases). A 2-Back task was administered as a measure for working memory and executive functioning. During this task, characters were presented on the computer screen after each other and participants had to react as fast as possible by pressing the spacebar if the character presented was the same as two characters before (i.e. the same character as one before the previous character). Each character was presented for 200 ms with an interval of 800 ms between two characters.

The mean reaction times of correct responses on the PVT, Go-NoGo and 2-Back tasks were used as dependent measures for performance in terms of speed. Note that the intervals between the beeps were shorter (between 1 and 5 s) in the PVT and Go-NoGo tasks during baseline phase to have a sufficient number of responses to calculate the mean reaction times as the tasks were shorter at baseline. In addition, percentage of correct responses was used as measure for accuracy on the Go-NoGo task and 2-Back tasks.

2.6.3. Physiological measures

Physiological arousal was investigated using heart rate, heart rate variability and skin conductance measures. These variables were measured continuously during the experiment using TMSi software. The electrodes for the Electrocardiography (ECG) measurements were placed at V1 and V6 using Kendall Arbo H124SG ECG electrodes. Electrodermal activity (EDA) was measured using two electrodes of which one was placed on the forefinger and one on the middle finger of the participant's non-dominant hand. A sampling frequency of 1024 Hz was used. ECG and EDA measurements of 5 min during the auditory PVT tasks (3-min PVT in baseline) were used for the analyses.

To minimize poor signal quality, artifacts in the ECG signals were removed following a procedure for EEG described by Van de Velde, Ghosh, and Cluitmans (1999). Time domain and frequency domain (Fast Fourier Transform) analyses were performed with Kubios analysis software (Niskanen, Tarvainen, Ranta-Aho, & Karjalainen,

2004) using adequate RR intervals. Mean heart rate (bpm) and the LF/HF power ratio (LF: .04 to. 15 Hz; HF: .15 to .40 Hz) were used as dependent measures in this study.

The EDA data during the PVT was downsampled to 32 Hz. Subsequently, a 1-Hz Butterworth low-pass filter was applied to determine the skin conductance level (SCL). The average level (μSiemens) during the measurement period was calculated and subsequently log-transformed due to the skewed distribution of the data.

2.6.4. Evaluation and beliefs of lighting condition

Subjective evaluation of the lighting conditions was assessed with eight 5-point semantic differential adjective items adopted from Flynn, Spencer, Martyniuk and Hendrik (1973). The subscale assessing experienced pleasantness of the lighting consisted of four items ('unpleasant – pleasant', 'uncomfortable – comfortable', 'disturbing – not disturbing' and 'causing glare – not causing glare') and was internally reliable with $\alpha = .86$. In addition, four single items assessed subjective evaluations concerning the color ('warm – cold'), brightness ('dim – bright') and distribution ('uniform – non-uniform') of the lighting and whether the lighting was activating ('relaxing – stimulating').

In addition, two separate items probed the adequacy of the amount and adequacy of the color of the lighting for a working environment. Visual comfort was measured with three items from Aries (2005; $\alpha = .45$). Due to the low reliability, we decided to not include this latter measure in the analyses.

Four semantic differential adjective items probed beliefs regarding the effects of the lighting conditions on work performance and mood. Three items concerned potential effects of the lighting on work performance ('unsuitable to work - suitable to work', 'makes it difficult to concentrate – makes it easy to concentrate', 'causes bad performance – cause good performance'; internal reliability of $\alpha = .83$) and one item related to whether the lighting affected their mood ('does not affect my mood – affects my mood').

2.6.5. Evaluation of the environment

Four items measured participants' overall impression of the environment. Three items ('beautiful', 'nice' and 'pleasant') were based on Staats, Kieviet, and Hartig (2003). The item 'comfortable' was added to this scale. Answers were given on a scale ranging from (1) 'not at all' to (7) 'extremely' ($\alpha = .90$). Attitude towards the working in the environment was measured with four items ('attractive', 'pleasant', 'positive' and 'agreeable') probing what it would be like to work in a comparable environment ($\alpha = .97$).

2.6.6. Potential confounding variables

Time awake, minutes of sleep, time spent outdoors, and traveling time outdoors prior to the experiment were assessed at the end of each session. Gender, age, light sensitivity, chronotype and global sleep quality were measured at the end of the third session, and trait vitality and general health were measured at the end of the fourth session. Subjective light sensitivity was measured with three items from Smolders, De Kort, and Cluitmans (2012; $\alpha = .70$). Chronotype was assessed with the Dutch translation of the Munich Chronotype questionnaire (MCTQ; Roenneberg et al., 2003) and computed according to the formula provided by Roenneberg et al. (2004). Global sleep quality was assessed with the Pittsburgh Sleep Quality Index (PSQI; Buysse, Reynolds, Monk, Berman, & Kupfer, 1989) consisting of 18 items concerning subjective sleep quality, sleep latency, sleep duration, sleep efficiency, sleep disturbances, sleeping medication and daytime dysfunction. General health was assessed with five items from the Dutch version of the SF-36 Health Survey (General health perception subscale, RAND-36; Van der Zee, Sanderman, Heyink, & de Haes, 1996; $\alpha = .69$). Trait subjective vitality was measured with the subjective trait level vitality scale (Ryan & Frederick, 1997; $\alpha = .79$).

2.6.7. Summary: overview of the independent and dependent variables

In the current study, we manipulated participants' mental state (fatigued vs. relaxed) and the illuminance level (1000 lx vs. 200 lx at the eye), to test the effects of bright light exposure on subjective experiences (alertness, mood, self-control capacity, appraisals and beliefs), task performance on a simple vigilance task (PVT) and two more complex tasks (Go-NoGo and 2-Back tasks), and physiological arousal of the autonomic nervous system (HR, HRV, SCL) under mental fatigue vs. a more rested state. We assessed these variables in two repeated measurement blocks (except for self-control capacity, appraisals and beliefs) during a half hour of light exposure, to investigate the development of the effects of the antecedent and lighting condition on alertness and arousal with time in session. In sum, we investigated alerting and activating effects of bright light exposure on the different dependent variables (i.e., self-report, performance and physiological measures) and tested whether these effects were moderated by participants' prior mental state and/or time in session. Moreover, we investigated whether appraisals and beliefs of the lighting differed between participants' fatigued vs. rested state, and reflected the effects on mental well-being and task performance.

2.7. Statistical analysis

Due to the hierarchical structure of the data and the fact that the dataset was incomplete, Linear Mixed Model (LMM) analyses were performed to analyze the results (separate analyses for each dependent variable). Linear mixed models differ from the generalized linear models in that the model contains fixed as well as random effects, and allow missing data. Mixed model analysis provides a flexible approach in situations where correlated data arise due to, for example, repeated measurements on each subject over time, because it allows to explicitly model the covariance structure of the data (Seltman, 2013).

To check for potential differences at baseline, LMM analyses were performed with Lighting condition (200 lx vs. 1000 lx at the eye) and Antecedent condition (Fatigue vs. Control) as fixed factors for the self-report, performance and physiological measures during the baseline phase. Then a manipulation check of the fatigue induction was performed using LMM analyses with Lighting condition and Antecedent condition as fixed factors and the subjective alertness and mood measures as dependent variables. Note that these self-report measures probed individuals' mental state immediately after the mental fatigue vs. control induction, but before the light treatment. To test the effects of illuminance level under mental fatigue or when rested, LMM analyses with Lighting condition, Antecedent condition and Measurement block (Block 1 vs. Block 2) as fixed factors were performed on the measures taken during the lighting conditions. Last, to test the effects of illuminance level and prior mental state on appraisals and beliefs about the lighting, LMM analyses with Lighting condition and Antecedent condition as fixed factors were performed.

In the analyses, Participant was added as independent random variable to group the data per participant, i.e. to indicate that the same participant was measured multiple times. To control for differences at baseline, the baseline measurement was added as covariate in the analyses, except for the LMM models for subjective self-control capacity, appraisals and beliefs – as these variables were measured only at the end of each session – and of course except for the baseline comparisons. In addition, Light sensitivity was added as covariate to control for individual differences in

subjective light sensitivity. Other person characteristics were not added to the model to avoid multicollinearity as they showed significant correlations with the other person characteristics or baseline measures. We decided to not add Time of day as additional factor to the model of these analyses as this was not the focus of the current study. Preparatory analyses did reveal some baseline differences in subjective and performance measures and therefore the corresponding baseline score was added as covariate in the hypothesis testing analyses. Note that potential differences in experiences, task performance and physiology between subjects participating in the morning vs. afternoon sessions may already be reflected in the baseline measures. In addition, we were particularly interested in the 3-way interaction effect of Lighting condition, Antecedent and Time of day, but these results should be considered with caution due to the relatively low number of measurements per cell (i.e. between 11 and 15).

Thus, the model used in the analyses for the experimental phase is a two-level model with Lighting condition, Antecedent condition, Measurement block and the corresponding baseline score (not for self-control, appraisals and beliefs) as within-subjects predictors for the dependent measures of each session (Level 1). Light sensitivity was added to the model as predictor for the dependent measures at the participant level (Level 2). Note that reported contrasts for interaction effects always refer to posthoc comparisons with Bonferroni correction.

3. Results

First, we investigated whether there were differences in sleep duration, time awake and time spent outside between the conditions. In addition, baseline comparisons for the dependent measures were performed to investigate whether differences in sleepiness, vitality, mood, task performance and physiological arousal already existed at baseline (prior to the manipulations). Subsequently, a manipulation check of the antecedent condition was performed to investigate whether participants felt indeed sleepier and more depleted immediately after the mental fatigue induction compared to the control condition. Next, the effects of the antecedent condition and illuminance at eye level on self-reported sleepiness, vitality, mood, and self-control capacity and on task performance and physiology were tested. Last, we explored whether evaluations of the lighting and the room and individuals' beliefs were dependent on the conditions.

3.1. Baseline comparisons

3.1.1. Confounding variables: sleep and time spent outside

Linear mixed model (LMM) analyses were performed to test whether there were differences the potential confounding variables, i.e., in sleep duration, minutes awake and minutes spent outside prior to the experiment, between the four experimental conditions. Results of these analyses revealed no significant main or interaction effects of Lighting condition and Antecedent condition on these three measures (all $p > .10$), suggesting that these variables did not significantly differ between the conditions.

3.1.2. Baseline self-reported sleepiness, vitality and mood

The results revealed no significant main effects of Lighting condition and Antecedent condition on the subjective measures at baseline (all $p > .10$). However, the interaction between Lighting condition and Antecedent condition did show significant baseline differences in vitality and positive affect at the start of the experiment [$F(1,75) = 8.09$; $p < .01$ and $F(1,77) = 5.73$; $p = .02$, respectively]. Participants felt more vital and reported higher positive affect prior to exposure to the 200 lx than to the 1000 lx condition at the start of the control sessions (both $p < .05$), but not at the start of the fatigue induction sessions (both $p > .10$; see Table 1). The interaction between Lighting condition and Antecedent condition on subjective sleepiness and negative affect approached significance [$F(1,76) = 3.00$; $p = .09$ and $F(1,77) = 3.82$; $p = .05$, respectively]. Posthoc comparisons revealed no significant differences between the 200 lx and 1000 lx in either the control or fatigue condition (all $p > .10$), but showed a non-significant trend for lower sleepiness and less negative affect in the control sessions compared to the fatigue sessions at baseline prior to the 200 lx conditions ($p = .06$ and $p = .05$, respectively), but not at the start of the 1000 lx conditions (both $p > .10$). The interaction effect between Lighting condition and Antecedent condition on tension was not significant ($F < 1$; *ns*), suggesting no significant baseline differences in self-reported tension.

3.1.3. Baseline performance

Results of the LMM analyses revealed no significant baseline differences for Lighting condition and Antecedent condition on the PVT mean reaction time (all $p > .10$). The results revealed also no significant baseline differences in speed or accuracy for Lighting condition or Antecedent condition on the Go-NoGo task (all $p > .10$), except for a significant interaction effect between Lighting condition and Antecedent condition on the percentage correct at baseline [$F(1,70) = 4.38$; $p = .04$]. Posthoc comparisons showed no significant differences between the 200 lx and 1000 lx in either the control or fatigue condition, but suggested a non-significant trend for a lower percentage correct prior to the 1000 lx fatigue sessions compared to the 1000 lx control sessions ($EMM_{Fatigue} = 92.42$; $SE = 1.06$; $EMM_{Control} = 94.69$; $SE = 1.08$; $p = .07$). Baseline comparisons for the 2-Back task measures revealed no significant main effect of Antecedent condition on the mean reaction time and percentage correct (both $p > .10$). There was, however, a significant baseline difference in percentage correct between the two lighting conditions [$F(1,76) = 5.59$; $p = .02$]. The percentage correct was lower prior to exposure to the 1000 lx condition than to the 200 lx condition ($EMM_{1000lx} = 77.44$; SE = 2.31; $EMM_{200lx} = 84.61$; SE = 2.28). The interaction effects between Lighting condition and Antecedent condition on speed and accuracy on the 2-Back task at baseline were not significant (both $p > .10$).

3.1.4. Baseline physiological arousal

The results for the baseline comparisons of heart rate, heart rate variability (HRV) and skin conductance level (SCL) revealed no significant main or interaction effects of Lighting condition and Antecedent condition (all $p > .10$). This suggests that there were no significant differences in physiological arousal between the conditions at the start of the sessions.

3.1.5. Summary of baseline comparisons

Overall, the baseline comparisons showed some differences in both self-report and performance measures at the start of the

Table 1
Estimated means for baseline comparisons: Self-report measures.

	Control				Fatigue			
	200 lx		1000 lx		200 lx		1000 lx	
	EMM	*SE*	*EMM*	*SE*	*EMM*	*SE*	*EMM*	*SE*
Sleepiness (KSS)	3.48	.31	4.14	.33	4.23	.31	3.93	.31
Vitality	**3.89**	**.13**	**3.48**	**.14**	3.52	.13	3.71	.13
Tension	2.29	.11	2.17	.11	2.32	.11	2.23	.12
Happy	**3.19**	**.16**	**2.82**	**.16**	3.00	.15	3.23	.16
Sad	1.20	.11	1.38	.12	1.47	.11	1.27	.11

Note. EMM stands for estimated marginal means and SE stands for standard errors. Significant differences are indicated in bold.

Table 2
Results of manipulation check: Self-report measures.

	Control		Fatigue		Statistics			
	EMM	*SE*	*EMM*	*SE*	*F*	*df*	*p*	R^2
Sleepiness (KSS)	**4.28**	**.22**	**6.12**	**.21**	**40.68**	**(1,75)**	**<.01**	**.44**
Vitality	**3.45**	**.10**	**2.77**	**.09**	**28.62**	**(1,75)**	**<.01**	**.45**
Tension	**2.08**	**.08**	**2.45**	**.07**	**14.10**	**(1,72)**	**<.01**	**.30**
Happy	**3.13**	**.08**	**2.41**	**.08**	**43.15**	**(1,74)**	**<.01**	**.45**
Sad	**1.28**	**.06**	**1.46**	**.06**	**5.01**	**(1,74)**	**.03**	**.39**

Note. EMM stands for estimated marginal means and SE stands for standard errors. Significant differences are indicated in bold. Pseudo *R*-Squared are given for the total mixed model at level 1, i.e. within sessions. This measure is the proportion of reduction in variance of residuals. Note that this measure can also have negative values (Raudenbush & Bryk, 2002).

sessions. Therefore, we decided, as a general rule, to add the corresponding baseline measure as covariate in subsequent analyses.

3.2. Manipulation check

Table 2 reports the results for the comparisons of the self-report measures between the mental fatigue vs. control condition immediately after the manipulation, yet before the lighting condition. The results showed that participants felt, as expected, sleepier and less energetic after the fatigue induction compared to the control condition (see Fig. 3). In addition, participants reported higher feelings of tension, less positive affect and more negative affect in the fatigue condition than in the control condition (see Table 2).

Lighting condition had no significant main effect on these subjective measures (all $F < 1$, *ns*). The effect of Antecedent condition on subjective sleepiness and vitality prior to the light treatment was not moderated by Lighting condition (both $F < 1$, *ns*). The interaction effect between Lighting condition and Antecedent condition on positive affect was, however, significant [$F(1,77) = 4.76$; $p = .03$], suggesting lower positive affect before exposure to 200 lx than to 1000 lx in the control conditions ($EMM_{200lx} = 2.96$; $SE = .11$; $EMM_{1000lx} = 3.30$; $SE = .12$; $p = .04$), but not in the fatigue conditions ($EMM_{200lx} = 2.48$; $SE = .11$; $EMM_{1000lx} = 2.33$; $SE = .11$; $p > .10$). In addition, there was a significant interaction effect between Lighting condition and Antecedent condition on feelings of tension [$F(1,72) = 4.93$; $p = .03$], suggesting a similar trend for more tension in the 200 lx control condition than in the 1000 lx control condition ($EMM_{200lx} = 2.21$; $SE = .10$; $EMM_{1000lx} = 1.96$; $SE = .11$; $p = .09$), but no significant difference in the fatigue condition ($EMM_{200lx} = 2.35$; $SE = .10$; $EMM_{1000lx} = 2.54$; $SE = .10$; $p = .17$). The interaction had no significant effect on negative affect ($F < 1$, *ns*).

3.2.1. Summary of results manipulation check

The fatigue induction was successful as participants reported increased feelings of sleepiness and lower feelings of vitality after the mental fatigue condition compared to the control condition. In addition, participants reported higher feelings of tension, less positive affect and more negative affect.

3.3. Effects on subjective indicators

In this section, we investigate the effect of Lighting condition and Antecedent condition on self-report indicators measured during exposure to a high vs. low illuminance level. Tables 3 and 4 show the statistics for the main effects of Antecedent condition and Lighting condition respectively, as well as the estimated marginal means for each variable per condition. The results revealed a significant main effect of Antecedent condition on subjective sleepiness and vitality, suggesting that the effect of this manipulation on these variables lasted also during the light exposure (see Table 3). Participants felt sleepier and less energetic during the light exposure after the mental fatigue induction than after the control condition. In addition, participants reported lower positive affect when fatigued than when they engaged in relaxing activities before the light exposure. There was no significant main effect of Antecedent condition on feelings of tension and sadness (see Table 3), showing that the differences that existed in these variables immediately after the fatigue vs. control induction leveled out during the light exposure when participants were engaging in the dependent tasks.

Lighting condition had also a significant main effect on subjective feelings of sleepiness and vitality: Participants reported, as expected, lower feelings of sleepiness and more vitality in the 1000 lx compared to the 200 lx condition (see Table 4). These results replicated earlier findings at night and during the day. For the current research question, we were particularly interested in the interaction between Lighting condition and Antecedent condition. This interaction approached significance for sleepiness [$F(1,179) = 3.62$; $p = .06$]. In line with our expectations, post-hoc comparisons showed a significant effect of Lighting condition on self-reported sleepiness only when participants experienced mental fatigue [$F(1,177) = 9.68$; $p < .01$], but not when rested ($F < 1$, *ns*; see Fig. 4). The interaction effect between Lighting condition and Antecedent condition on vitality was not significant ($F < 1$, *ns*), suggesting that the effect of lighting on vitality was not significantly moderated by mental fatigue, although differences between the control and fatigue conditions during the experimental lighting conditions appeared smaller in the 1000 lx condition than in the 200 lx condition (see Fig. 4). Potentially this effect is masked due to the baseline difference that was present for vitality.

Lighting condition also had a significant effect on positive affect ($p < .01$): Participants felt happier in the 1000 lx condition than in the 200 lx condition (see Table 4). This effect was not moderated by Antecedent condition [$F(1,179) = 2.21$; $p = .14$]. The effect of Lighting condition on tension approached significance suggesting a non-significant trend for lower tension in the 1000 lx condition

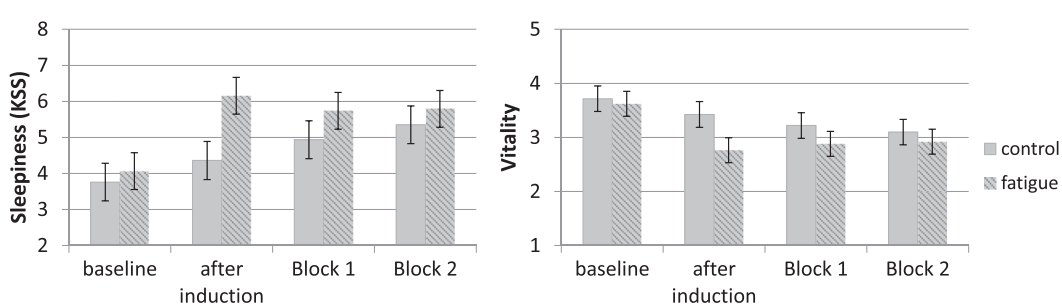

Fig. 3. Development of sleepiness (left) and vitality (right) over time in session.

Table 3
Results on the subjective and objective indicators for Fatigue induction during the light exposure.

	Control		Fatigue		Statistics			
	EMM	*SE*	*EMM*	*SE*	*F*	*df*	*p*	R^2
Subjective measures								
Sleepiness (KSS)	**5.02**	**.24**	**5.66**	**.24**	**14.47**	**(1,177)**	**<.01**	**.28**
Vitality	**3.21**	**.11**	**2.95**	**.10**	**15.57**	**(1,177)**	**<.01**	**.38**
Tension	2.36	.07	2.45	.06	2.39	(1,173)	.12	.11
Happy	**2.86**	**.10**	**2.62**	**.09**	**11.72**	**(1,176)**	**<.01**	**.31**
Sad	1.38	.06	1.39	.06	.09	(1,165)	.77	.30
Lack of self-control	**3.55**	**.26**	**4.00**	**.26**	**5.09**	**(1,74)**	**.03**	**.13**
Performance measures								
PVT								
Mean RTs	**342.15**	**6.51**	**357.24**	**6.46**	**11.61**	**(1,166)**	**<.01**	**.57**
Go-NoGo								
Mean RTs	**451.50**	**10.81**	**478.75**	**10.73**	**14.62**	**(1,169)**	**<.01**	**.39**
Percentage correct	95.10	.48	95.47	.48	.66	(1,163)	.42	.04
2-Back task								
Mean RTs	**565.50**	**10.30**	**581.40**	**10.17**	**5.46**	**(1,167)**	**.02**	**.06**
Percentage correct	82.68	1.49	81.22	1.46	1.38	(1,174)	.24	.20
Physiological arousal								
ECG								
Heart rate (bpm)	73.75	.74	73.13	.75	1.27	(1,121)	.26	.63
HRV (lf/hf ratio)	2.44	.28	2.49	.29	.05	(1,128)	.83	.12
EDA								
SCL (lg μSiemens)	.43	.04	.43	.04	.02	(1,123)	.90	.68

Note. EMM stands for estimated marginal means and SE stands for standard errors. Significant differences are indicated in bold. Pseudo R-Squared are given for the total mixed model at level 1, i.e. within sessions. This measure is the proportion of reduction in variance of residuals. Note that this measure can also have negative values (Raudenbush & Bryk, 2002). Please note also that lack of self-control, in contrast to the other measures, was measured only once at the end of each session.

than in the 200 lx condition. Lighting condition had no significant main effect on negative affect (see Table 4). No significant interaction effects between Lighting condition and Antecedent condition emerged for tension and negative affect (both $p > .10$).

Table 4
Results on the subjective and objective indicators for Lighting condition.

	200 lx		1000 lx		Statistics			
	EMM	*SE*	*EMM*	*SE*	*F*	*df*	*p*	R^2
Subjective measures								
Sleepiness (KSS)	**5.54**	**.24**	**5.14**	**.24**	**5.46**	**(1,179)**	**.02**	**.28**
Vitality	**2.98**	**.10**	**3.19**	**.11**	**10.25**	**(1,179)**	**<.01**	**.38**
Tension	2.46	.06	2.35	.07	3.15	(1,176)	.08	.11
Happy	**2.60**	**.09**	**2.88**	**.10**	**14.94**	**(1,177)**	**<.01**	**.31**
Sad	1.42	.06	1.35	.06	2.09	(1,166)	.15	.30
Lack of self-control	3.80	.26	3.75	.26	.06	(1,75)	.81	.13
Performance measures								
Auditory PVT								
Mean RTs	350.83	6.44	348.56	6.53	.26	(1,169)	.61	.57
Auditory Go-NoGo								
Mean RTs	**455.80**	**10.75**	**474.46**	**10.80**	**6.85**	**(1,170)**	**<.01**	**.39**
Percentage correct	95.59	.48	94.97	.49	.66	(1,163)	.42	.04
2-Back task								
Mean RTs	571.87	10.22	575.03	10.27	.21	(1,169)	.65	.06
Percentage correct	**83.80**	**1.46**	**80.10**	**1.49**	**8.71**	**(1,175)**	**<.01**	**.20**
Physiological arousal								
ECG								
Heart rate (bpm)	73.34	.74	73.54	.76	.14	(1,124)	.71	.63
HRV (lf/hf ratio)	2.56	.28	2.38	.29	.65	(1,127)	.42	.12
EDA								
SCL (lg μSiemens)	**.41**	**.04**	**.45**	**.04**	**7.45**	**(1,125)**	**<.01**	**.68**

Note. EMM stands for estimated marginal means and SE stands for standard errors. Significant differences are indicated in bold. Pseudo *R*-Squared are given for the total mixed model at level 1, i.e. within sessions. This measure is the proportion of reduction in variance of residuals. Note that this measure can also have negative values (Raudenbush & Bryk, 2002). Please note also that lack of self-control, in contrast to the other measures, was measured only once at the end of each session.

Measurement block had no significant main or interaction effects on the self-report measures for sleepiness and mood (all $p > .10$). Thus the effects of Antecedent condition and Lighting condition were not moderated by time in session, suggesting that the reported effects of mental fatigue and illuminance level were immediate and consistent during the light exposure. Light sensitivity was no significant predictor for sleepiness, vitality, positive or negative affect (all $p > .10$), but was significantly related to tension [$F(1,23) = 11.29$; $p < .01$] suggesting that participants who reported a higher light sensitivity experienced more tension during the light exposure ($B = .27$). The corresponding baseline scores of the dependent measures were all significantly and positively correlated with the subjective measures during the light treatment (all $p < .01$).

Lack of self-control capacity was measured only once, at the end of each session. An LMM analysis with Lighting condition and Antecedent condition as fixed factors and Light sensitivity as covariate revealed a significant main effect of Antecedent condition with a higher lack of self-control after the fatigue induction than at the end of the control sessions (see Table 3). Lighting condition had no significant main effect on lack of self-control (see Table 4), but the interaction between Lighting condition and Antecedent condition was significant [$F(1,74) = 6.06$; $p = .02$]. This interaction showed that the difference between the fatigue and control conditions was only significant in the 200 lx condition ($EMM_{fatigue} = 4.27$; $SE = .29$; $EMM_{control} = 3.33$; $SE = .29$; $p < .01$), and not in the 1000 lx condition ($EMM_{fatigue} = 3.73$; $SE = .29$; $EMM_{control} = 3.77$; $SE = .30$; $p > .10$). Light sensitivity was no significant predictor for lack of self-control ($F < 1$, *ns*).

3.3.1. Summary lighting effects on subjective indicators

Participants reported lower sleepiness, higher vitality and more positive affect in the 1000 lx vs. 200 lx condition. The effect of illuminance level on subjective sleepiness occurred particularly among fatigued participants. In addition, the effect of the fatigue induction on self-control capacity at the end of the session was most pronounced in the 200 lx condition.

3.4. Effects on performance

Comparable to the results on subjective sleepiness and subjective self-control capacity, we expected effects of illuminance level on performance indicators to be moderated by the participant's prior mental state. Our hypothesis was that Lighting condition would show immediate effects on performance indicators for fatigued participants, yet smaller and/or delayed effects for rested participants. Results of the main effects of Antecedent condition and Lighting condition on the speed and accuracy performance measures are reported in Tables 3 and 4 respectively.

3.4.1. Auditory PVT: effects on sustained attention

Results revealed a significant main effect of the Antecedent condition on the auditory PVT mean reaction time: Participants had longer reaction times when fatigued than when rested (see Table 3). This suggests that the fatigue induction also resulted in decrements in speed on the PVT during the light treatment. Measurement block had a significant main effect on the mean reaction time [$F(1,165) = 17.65$; $p < .01$], indicating a declining performance from Block 1 to Block 2 ($EMM_{Block1} = 340.49$; $SE = 6.47$; $EMM_{Block2} = 358.90$; $SE = 6.48$). The interaction between Measurement block and Antecedent condition was not significant ($F < 1$, *ns*), suggesting that reaction times increased with time in session for both fatigued and rested participants and that fatigued participants performed worse on the auditory PVT than rested

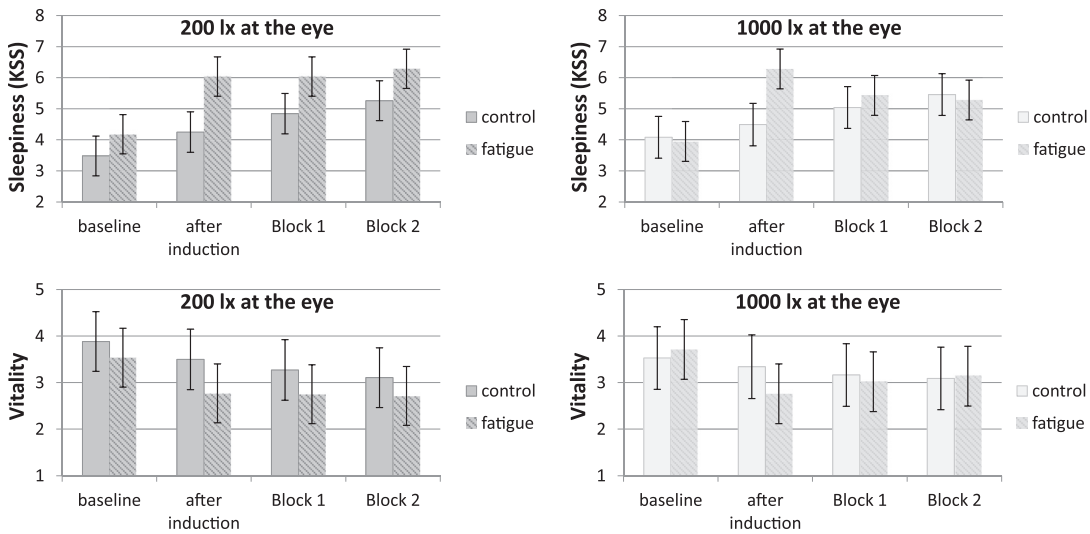

Fig. 4. Development sleepiness (upper row) and vitality (lower row) over time in session in 200 lx (left) and 1000 lx (right) condition for fatigued vs. rested participants.

participants regardless of time in session after the fatigue vs. control induction.

Lighting condition, however, had no significant main effect on the mean reaction time (see Table 4). In contrast to our hypothesis, the interaction between Lighting condition and Antecedent condition on the mean reaction time was also not significant ($F < 1$; *ns*), suggesting no state-dependent effect of bright light exposure on sustained attention. In contrast, the interaction between Lighting condition and Measurement block on the PVT mean reaction time showed a non-significant trend [$F(1,165) = 3.65$; $p = .06$], showing a significant increase in reaction times with time in session for the 200 lx condition ($EMM_{Block1} = 337.45$; $SE = 7.10$; $EMM_{Block2} = 364.20$; $SE = 7.13$; $p < .01$), but not for the 1000 lx condition ($EMM_{Block1} = 343.52$; $SE = 7.25$; $EMM_{Block2} = 353.60$; $SE = 7.25$; $F(1,165) = 2.54$; $p = .11$). This suggests that the deterioration over time in reaction time was less pronounced in the 1000 lx condition compared to the 200 lx condition (see Fig. 5). The 3-way interaction between Lighting condition, Measurement block and Antecedent condition was not significant ($F < 1$, *ns*). The baseline PVT mean reaction time was a significant predictor for the mean reaction time during the light exposure [$B = .87$; $F(1,167) = 264.89$; $p < .01$]. Light sensitivity was not significantly related to the mean reaction time on the auditory PVT during the lighting conditions ($F < 1$; *ns*).

3.4.2. Auditory Go-NoGo task: effects on inhibitory capacity

Results of the LMM analysis for the auditory Go-NoGo task revealed a significant main effect of Antecedent condition on the mean reaction time, but not on the percentage of correct responses. Participants' responses to targets during the light treatment were slower after the mental fatigue induction than after the control condition (see Table 3). Lighting condition had a significant main effect on mean reaction time with, in contrast to our expectations, longer reaction times in the 1000 lx condition compared to the 200 lx condition (see Table 4). The interaction between Lighting condition and Antecedent condition was not significant for the mean reaction time ($F < 1$; *ns*), suggesting that the effect of illuminance level on speed was not moderated by mental fatigue. In contrast, Lighting condition had no significant main effect on the percentage correct, but its interaction with Antecedent condition was significant for the percentage of correct responses [$F(1,167) = 11.32$; $p < .01$]. Participants had a higher percentage correct in the 200 lx condition than in the 1000 lx condition when they engaged in relaxing activities before the light treatment ($EMM_{200lx} = 96.20$; $SE = .58$; $EMM_{1000lx} = 93.99$; $SE = .59$; $F(1,167) = 11.40$; $p < .01$), while after the fatigue induction the pattern appeared reversed, but was not significantly different ($EMM_{200lx} = 94.99$; $SE = .58$; $EMM_{1000lx} = 95.95$; $SE = .59$; $p = .14$). Note also that in the 1000 lx condition participants had a higher percentage correct after the fatigue induction than after the control condition [$F(1,168) = 8.55$; $p < .01$]. In the 200 lx condition, there was a non-significant trend for the opposite, and expected, pattern [$F(1,163) = 3.50$; $p = .06$] with a lower percentage correct after the fatigue induction compared to the control condition (see Fig. 6).

Measurement block had a significant main effect on the mean reaction time in the auditory Go-NoGo task [$F(1,167) = 7.23$; $p < .01$], with increasing reaction times from Block 1 to Block 2 ($EMM_{Block1} = 455.75$; $SE = 10.74$; $EMM_{Block2} = 474.51$; $SE = 10.76$). In addition, the effect of Measurement block on percentage correct showed a non-significant trend for a decrease in accuracy from Block 1 to Block 2 ($EMM_{Block1} = 95.67$; $SE = .48$; $EMM_{Block2} = 94.90$; $SE = .48$; $F(1,158) = 2.88$; $p = .09$). The interaction effects of

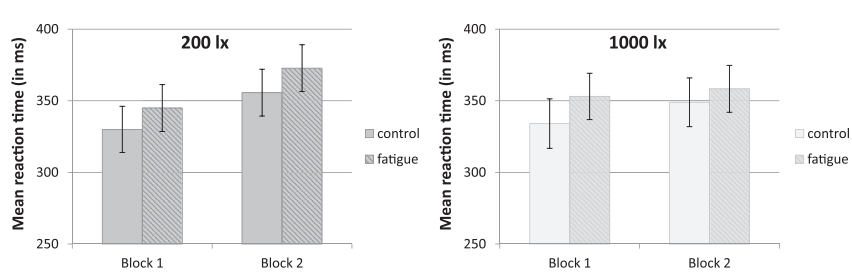

Fig. 5. Mean reaction time auditory PVT in 200 lx (left) and 1000 lx (right) condition in measurement block 1 and block 2 for fatigued vs. rested participants.

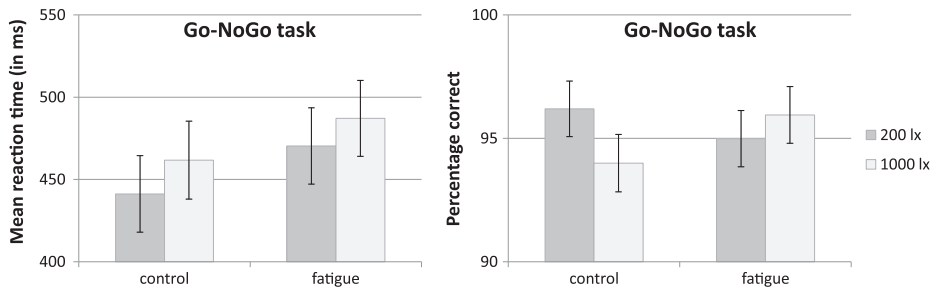

Fig. 6. Mean reaction times (left) and percentage correct (right) on auditory Go-NoGo task for fatigued vs. rested persons in the 200 lx and 1000 lx condition.

Measurement block with Lighting condition or with Antecedent condition were not significant (all $p > .10$), suggesting that the effects of illuminance level or mental fatigue on speed and accuracy on the auditory Go-NoGo task did not change with duration of the light treatment. Light sensitivity was not significantly related to performance on the auditory Go-NoGo task (both $p > .10$). The baseline mean reaction time was a significant predictor for the mean reaction time during the light exposure [$B = .72$; $F(1,195) = 95.38$; $p < .01$], while the baseline percentage correct was not significantly related to accuracy on the auditory Go-NoGo task during the light exposure [$B = .09$; $F(1,171) = 2.72$; $p = .10$].

3.4.3. 2-Back task: effects on ability to stay focused and working memory

Comparable to the results on the auditory Go-NoGo task, Antecedent condition had a significant main effect on the visual 2-Back task mean reaction time, but not on the percentage of correct responses. Fatigued participants responded slower to target characters than rested participants (see Table 3). Lighting condition had no significant main effect on the mean reaction time, but had a significant effect on the percentage correct responses with, in contrast to our expectation, a lower percentage correct in the 1000 lx conditions than in the 200 lx conditions (see Table 4). This effect of Lighting condition on percentage correct targets was not moderated by Antecedent condition ($F < 1$; *ns*). The interaction between Lighting condition and Antecedent condition was also not significant for the mean reaction time ($F < 1$; *ns*).

Measurement block had no significant effects on the 2-Back measures (all $F < 1$; *ns*), except for a non-significant trend for an interaction effect between Antecedent condition and Measurement block on the mean reaction time [$F(1,166) = 2.96$; $p = .09$]. Posthoc comparisons showed that the difference in speed between fatigued participants and rested participants was only significant in the second measurement block ($EMM_{Fatigue} = 586.90$; $SE = 11.17$; $EMM_{Control} = 559.41$; $SE = 11.40$; $p < .01$), but not in the first measurement block ($EMM_{Fatigue} = 575.90$; $SE = 11.18$; $EMM_{Control} = 571.59$; $SE = 11.40$; $p > .10$). While participants became faster in the control condition, reaction times increased in the fatigue condition.

The baseline mean reaction time was a significant predictor for the mean reaction time during the light exposure [$B = .12$; $F(1,197) = 6.06$; $p < .01$]. In addition, the baseline percentage correct was a significant predictor for accuracy on the 2-Back task during the light treatment [$B = .21$; $F(1,199) = 27.36$; $p < .01$]. Light sensitivity was not related to performance on the 2-Back task (both $p > .10$).

3.4.4. Summary of effects on task performance

The different performance indicators showed no stronger or more immediate effects when participants suffered from mental fatigue. While the results on the PVT suggested that sustained attention deteriorated less over time in the 1000 lx condition compared to the 200 lx condition, results on the Go-NoGo task and 2-Back task revealed some performance-undermining effects of bright light.

3.5. Effects on physiological indicators

LMM analyses for heart rate, HRV and SCL showed no significant differences in physiological arousal measured with ECG and EDA during the light exposure between fatigued and rested participants (see Table 3). In addition, there were no significant main or interaction effects of Lighting condition on heart rate and HRV (all $p > .10$). In contrast, Lighting condition had a significant main effect on EDA with a higher skin conductance level in the 1000 lx condition than in the 200 lx condition (see Table 4). The interaction between Lighting condition and Antecedent condition on SCL was also significant [$F(1,124) = 12.52$; $p < .01$]. Posthoc comparisons showed a higher skin conductance level when exposed to 1000 lx than to 200 lx in the control condition ($EMM_{1000lx} = .48$; $SE = .04$; $EMM_{200lx} = .38$; $SE = .04$; $p < .01$), but no difference between the lighting conditions for fatigued participants; their skin conductance level lay in between ($EMM_{1000lx} = .42$; $SE = .04$; $EMM_{200lx} = .44$; $SE = .04$; $p > .10$). Measurement block had no significant main or interaction effects on the physiological arousal measures (all $p > .10$). The corresponding baseline measures were significant predictors for heart rate, HRV and SCL during the light treatment ($B = .64$, $B = .28$ and $B = .84$ respectively, all $p < .01$).

3.5.1. Summary effects physiological indicators

Lighting condition had no significant effects on heart rate or heart rate variability. Bright light exposure induced a higher skin conductance level, yet only in the control condition and not when participants were fatigued prior to the light exposure. Together these results suggest no more pronounced effects of exposure to a higher illuminance level on autonomic nervous activity.

3.6. Subjective evaluations and beliefs

LMM analyses were performed to investigate the effect of Lighting condition and Antecedent condition on the subjective appraisals and beliefs concerning the lighting and the evaluation of the environment. In Table 5, the *F*-statistics of the LMM analyses on the evaluations and beliefs for Lighting condition are reported. In addition, the estimated marginal means for these variables in both the 200 lx and 1000 lx conditions are reported in Table 5.

3.6.1. Subjective appraisals

Antecedent condition had no significant main effects on the subjective appraisals of the lighting (all $p > .10$), suggesting that participants' evaluations did not differ between the fatigue and control condition. Lighting condition had a significant main effect

Table 5
Results on evaluations and beliefs.

	200 lx		1000 lx		Statistics			
	EMM	*SE*	*EMM*	*SE*	*F*	*df*	*p*	R^2
Subjective appraisals								
Pleasantness	**2.97**	**.16**	**2.36**	**.17**	**13.61**	**(1,71)**	**<.01**	**.24**
Activating	**3.08**	**.14**	**3.60**	**.15**	**12.18**	**(1,73)**	**<.01**	**.15**
Color	3.42	.19	3.34	.19	.23	(1,72)	.64	-.02
Brightness	**3.43**	**.14**	**4.52**	**.15**	**42.75**	**(1,72)**	**<.01**	**.40**
Distribution of lighting	2.63	.23	2.60	.23	.03	(1,71)	.87	.03
Adequacy amount of light	2.78	.18	2.57	.18	1.33	(1,72)	.25	.06
Adequacy color of light	3.27	.17	3.42	.18	.84	(1,71)	.36	.01
Beliefs								
Effect on work performance	2.99	.14	2.98	.15	<.01	(1,72)	.97	.05
Effect on mood	**2.98**	**.19**	**3.48**	**.19**	**8.85**	**(1,70)**	**<.01**	**.12**
Evaluations environment								
General evaluation	2.80	.18	2.78	.18	.02	(1,69)	.90	.02
Attitude working	2.44	.22	2.71	.23	3.80	(1,69)	.06	.08

Note. EMM stands for estimated marginal means and SE stands for standard errors. Significant differences are indicated in bold. Pseudo R-Squared are given for the total mixed model at level 1, i.e. within sessions. This measure is the proportion of reduction in variance of residuals. Note that this measure can also have negative values (Raudenbush & Bryk, 2002).

on pleasantness and brightness of the lighting, and whether the lighting was perceived as activating (see Table 5). The results revealed that the lighting in the 1000 lx (at the eye) condition was experienced as brighter and more activating than in the 200 lx (at the eye) condition. Participants evaluated the 1000 lx condition, however, as less pleasant than the 200 lx lighting condition. This effect on experienced pleasantness was moderated by Antecedent condition, as indicated by a significant interaction between Lighting condition and Antecedent condition [$F(1,70) = 4.21$; $p = .04$]. Posthoc comparisons showed that the 1000 lx condition was only experienced as less pleasant than the 200 lx condition in the control condition ($EMM_{1000lx} = 3.03$; $SE = .20$; $EMM_{200lx} = 2.08$; $SE = .21$; $p < .01$), and not after the fatigue condition ($EMM_{1000lx} = 2.64$; $SE = .20$; $EMM_{200lx} = 2.91$; $SE = .19$; $p = .24$).

The evaluations concerning the color and distribution of the lighting did not differ between conditions (all $p > .10$). There was no significant main effect of Lighting condition on the experienced adequacy of the lighting for office spaces in terms of intensity or color. The interaction effect between Lighting condition and Antecedent condition on the adequacy of the light intensity was, however, significant [$F(1,71) = 4.34$; $p = .04$]. Rested participants evaluated the 1000 lx condition as less adequate for office environments compared to the 200 lx condition ($EMM_{1000lx} = 2.37$; $SE = .23$; $EMM_{200lx} = 2.97$; $SE = .22$; $p = .03$). The opposite pattern appeared after the fatigue induction, but this difference was not significant ($EMM_{1000lx} = 2.76$; $SE = .22$; $EMM_{200lx} = 2.59$; $SE = .21$; $p > .10$). Participants' light sensitivity was not significantly related to their appraisals (all $p > .10$).

3.6.2. Beliefs

LMM analyses were also performed to explore whether participants' beliefs concerning the effect of the lighting condition on performance and mood differed between the conditions. Antecedent condition had no significant main effect on these beliefs (both $p > .10$). There was also no significant main effect of Lighting condition on the beliefs about the effect of the lighting on performance, but there was a significant main effect of Lighting condition on the beliefs concerning the effect of the lighting on mood (see Table 5). Participants indicated that they believed that the lighting condition affected their mood more in the 1000 lx (at the eye) condition than in the 200 lx (at the eye) condition. The interaction effects between Lighting condition and Antecedent condition on the beliefs were not significant (both $p > .10$). This suggests that the difference in beliefs about effects on mood was not more pronounced after the fatigue induction vs. the control condition. Light sensitivity was also not significantly related to the beliefs (both $F < 1$; *ns*).

3.6.3. Evaluation of the environment

There were no significant main or interaction effects of Lighting condition and Antecedent condition on the general evaluation of the room (all $F < 1$; *ns*). In addition, Antecedent condition did not have an effect on participants' attitude towards working in a comparable environment ($F < 1$, *ns*). There was, however, a non-significant trend for a more positive attitude towards working in a comparable environment in the 1000 lx compared to the 200 lx condition (see Table 5). The interaction between Lighting condition and Antecedent condition also approached significance [$F(1,69) = 3.24$; $p = .08$]. Posthoc comparisons showed that participants had a more positive attitude towards working in a comparable environment with the 1000 lx condition than with the 200 lx when they were fatigued before the light treatment ($EMM_{1000lx} = 2.83$; $SE = .25$; $EMM_{200lx} = 2.30$; $SE = .24$; $p < .01$). There was no difference in attitude between the two lighting conditions for the rested participants ($EMM_{1000lx} = 2.59$; $SE = .25$; $EMM_{200lx} = 2.57$; $SE = .24$; $p > .10$). Light sensitivity was no significant predictor for these subjective evaluations of the room (both $F < 1$; *ns*).

3.6.4. Summary appraisals and beliefs concerning the lighting condition

Participants evaluated the 1000 lx at the eye condition as brighter and more activating and reported stronger beliefs concerning the effect of the higher illuminance level on mood. While these effects were not moderated by mental fatigue, participants evaluated the bright light condition as less pleasant and less adequate when rested, but had a more positive attitude towards working in a comparable environment under 1000 lx when fatigued.

4. Discussion

The current study investigated whether effects of bright light exposure during daytime on feelings, task performance, physiological arousal and appraisals are more pronounced when persons suffer from mental fatigue, not from sleep pressure but due to mental resource depletion. Results showed that the effects of illuminance level on subjective sleepiness and self-control capacity were indeed dependent on individuals' psychological state. In line with our expectations, exposure to bright light had an effect on subjective sleepiness mainly when participants felt fatigued, effectively replenishing them close to the level of rested participants. A similar pattern was shown for subjective vitality, yet the effect of exposure to a higher illuminance level on vitality was not significantly moderated by persons' state, potentially because a baseline difference masked expected effects. Differences in subjective self-control capacity between fatigued and rested participants occurred only in the 200 lx light exposure. There was no 'self-control gap' between fatigued and rested participants who had experienced 1000 lx at the eye, in other words, the bright light appeared to have replenished perceived self-control capacity of participants when suffering from mental fatigue and resource depletion.

Although these subjective indicators showed a quite consistent pattern in line with our hypotheses, results on the performance measures showed mixed results. While performance on a simple reaction time task benefited from bright light exposure,

performance on the other more complex tasks appeared to be affected adversely by brighter light. Results on the auditory Psychomotor Vigilance Task (PVT) showed a larger increase in response times with time in session in the 200 lx condition than in the 1000 lx condition, suggesting that the effect of bright light on sustained attention was – in line with the results of Smolders, De Kort, & Cluitmans (2012) – dependent on duration of exposure. This delayed effect of bright light on performance occurred regardless of the induction (fatigue vs. control) prior to the light treatment. From the fact that the effect of bright light was not moderated by the fatigue induction, it appears unlikely that this delayed effect of daytime exposure to a higher illuminance level can be attributed to increasing fatigue of the participants. In that case we should have seen a more immediate and/or stronger effect of bright light when participants were fatigued prior to the light exposure in the current study. Effects on the Go-NoGo task and 2-Back task, probing inhibitory capacity and short-term memory respectively, were also not dependent on participants' prior state, except for a lower accuracy on the Go-NoGo task in the 1000 lx vs. 200 lx condition only when participants experienced a more relaxed state. Speed on the Go-NoGo task and accuracy on the 2-Back task revealed performance-reducing effects, regardless of whether participants experienced resource depletion or not and independent of the duration of exposure.

Together, the performance measures showed no stronger or more immediate effects after the mental fatigue induction, and results even revealed some adverse effects of bright light on the more complex cognitive tasks. It is unlikely that the effects on cognitive task performance are due to a low visibility of the task in the bright light condition as we see performance-undermining effects on both the visual 2-Back task as well as on the auditory Go-NoGo task. Moreover, other studies have shown increments in performance on visual cognitive tasks but not on sustained attention tasks when exposed to bright light both during daytime after hours of exposure to dim light (Rüger et al., 2005) or at night (Boyce et al., 1997; Kretschmer, Schmidt, & Griefahn, 2012). In contrast, exposure to a higher CCT in the evening enhanced speed on an auditory PVT and visual Go-NoGo task, while it did not affect performance on a visual executive functioning task (Chellappa et al., 2011). Although these studies did not report performance-undermining effects of exposure to more intense or more bluish white light, these results do also suggest that effects of light on task performance depend on the type of task. This is particularly relevant, as numerous studies in the past have only employed a single task – typically a vigilance task – perhaps implicitly overestimating the effects of bright light on performance in general. In fact, the current results revealed only beneficial alerting effects of bright light exposure on performance in a sustained attention task, and not on the other indicators for cognitive performance during regular daytime hours. As work-related as well as other daily tasks often requires more than sustained attention, this calls for further research to the effects of illuminance level on different aspects of human performance.

At this moment we can only speculate on the reasons for the differential effects on task performance in our study. A possible explanation is perhaps offered by arousal-level theories which would predict better performance on simple tasks under high – bright light – stimulation, yet worse performance on more complex tasks (Anderson, 1994; Yerkes & Dodson, 1908). This would imply that there exists an optimal illuminance level to induce arousal to engage in certain tasks based on the difficulty of the task (see also CIE, 2004/2009). As we employed only two light manipulations (200 lx vs. 1000 lx at the eye) and a limited set of performance tasks, future research is necessary to determine the optimal illuminance level for different types of performance tasks. It should be noted however, that a review by Veitch (2001) suggested that earlier research did not provide clear evidence for an inverted-U shaped relation between illuminance level and visual task performance as a function of task difficulty. In addition, some tasks may be more sensitive to a light manipulation than others, not only based on their complexity, but also on the required cognitive processes and skills to perform the task.

In the current study, participants always first engaged in the PVT and then in the Go-NoGo and 2-back tasks, without a rest period in between tasks. Task engagement on a first performance task may affect performance on a subsequent task that requires effort (e.g., Baumeister, Bratslavsky, Muraven, & Tice, 1998; Inzlicht & Schmeichel, 2012; Kaplan & Berman, 2010; Meijman & Mulder, 1998). It is, however, unlikely that participants were less motivated and put less effort in the cognitive performance tasks after performing better on the sustained attention task in the 1000 lx condition as the effect on the PVT was most pronounced in the second measurement block while the performance-undermining effect on the Go-NoGo and 2-Back tasks occurred also in the first block. To investigate effects of bright light on performance as a function of different types of tasks and eliminate possible order effects, future research could assess lighting effects on different performance tasks with sufficient rest periods in between the tasks.

Physiological indicators also revealed mixed results for the effects of illuminance level on autonomic nervous activity. Heart rate and heart rate variability (HRV) revealed no significant main or state-dependent effects of bright light. In line with earlier results (Smolders, 2013), there was an effect of lighting condition on skin conductance level with a higher level in the 1000 lx vs. 200 lx (at the eye) condition. Although skin conductance level (SCL) did show an arousal response in rested participants when exposed to bright light in the current study, this did not appear for fatigued persons. The current results on heart rate contrast findings by Smolders, De Kort, & Cluitmans (2012), but are in line with the results by Rüger et al. (2006) who also reported no measurable effect of illuminance level on heart rate during daytime. Thus, more research is necessary to establish potential effects of light on autonomic nervous activity during daytime.

The self-reports, task performance and physiological measures were administered in two repeated measurement blocks with a relatively large sample size, at least compared to earlier studies (e.g., Badia et al., 1991; Phipps-Nelson et al., 2003; Rüger et al., 2006), to investigate potential state-dependent effects of bright light exposure on alertness and arousal during daytime. Unfortunately, some measures revealed differences at baseline despite the fact that we paid special attention to minimize differences prior to the experimental conditions. In all analyses, we therefore controlled for participant's baseline scores of the corresponding session. Note that we employed a within-subjects design and asked persons to participate four times at the same time of day, to control for potential differences in alertness and arousal between sessions due to circadian variations in sleep pressure. Moreover, there were no significant differences in sleep pressure due to homeostatic regulation between the conditions as participants reported no differences in sleep duration or time awake. The baseline differences can therefore not be attributed to circadian or homeostatic modulation.

4.1. Potential underlying mechanism

In the current study, we did not employ EEG or fMRI to measure temporal or spatial modulations in brain activity during the light exposure. Due to different potential pathways and brain areas that can be activated by light simultaneously (Berson, Dunn, & Takao, 2002; Hattar et al., 2006; Vandewalle et al., 2009), we cannot

determine the neural processes underlying the effects found in the current study. Yet, the current study did shed some light on potential psychological mechanism as we also measured participants' mood and their appraisals and beliefs concerning the lighting.

Results showed that illuminance level had a significant effect on positive affect with participants reporting more positive affect in the 1000 lx condition than in the 200 lx (at the eye) condition, regardless of their prior mental state or time in session. This contrasts the state-dependent effects of exposure to a higher illuminance level on subjective sleepiness and self-control capacity as well as the duration-dependent effect on sustained attention, allowing us to discard positive affect as the underlying mechanism for the alerting effects reported here. Moreover, the fact that participants reported a higher positive affect in the 1000 lx condition also indicates that participants did not perform better on the Go-NoGo task and 2-Back task in the 200 lx condition due to a more positive mood.

Subjective evaluations of the lighting revealed that the participants rated the two lighting conditions differently. They experienced the 1000 lx (at the eye) condition as more bright and activating, regardless of their prior mental state. The 1000 lx condition was also experienced as less pleasant and less adequate for working when participants were more relaxed, but not when persons suffered from mental fatigue and resource depletion. In contrast, fatigued participants had a more positive attitude towards working in a comparable environment with brighter light. These results suggest that persons may appreciate a higher illuminance level more – or at least not less – than regular office lighting when they suffer from mental fatigue. These appraisals can therefore also not explain why both fatigued and rested participants performed better on the cognitive performance tasks in the 200 lx (at the eye) condition.

Generally, participants reported that they believed that the lighting conditions affected their mood and performance level, at least to some extent. In line with the mixed results on the performance measures, there were no significant differences in beliefs concerning the effect of lighting on performance between the lighting conditions. In contrast, beliefs that the lighting condition affected their mood were stronger in the 1000 lx than in the 200 lx condition – which was indeed in line with the effects on mood. It should be noted, however, that participants' beliefs were always asked at the end of the session, i.e., after the subjective mood and task performance measures during the light exposure. Therefore, participants' beliefs may have been affected by the reflection on their mood and experienced performance level during the light treatment. Future research is required to establish whether persons' beliefs and expectations about potential effects of a certain light condition on their mood and performance moderate light's effects on affective and cognitive functioning.

4.2. Conclusion and implications for future research

We compared the effect of bright light on subjective and objective indicators of alertness and arousal under different mental states. The results showed state-dependent effects of bright light on subjective sleepiness and self-control capacity, while the effects on task performance and autonomic nervous activity measures were not more pronounced under mental fatigue. To our knowledge, we are the first who tested the effects of light on mental wellbeing and performance of healthy individuals as a function of their antecedent psychological state, independent on time of day. Complementing earlier studies performed among persons suffering from sleepiness and fatigue due to high circadian and/or homeostatic sleep pressure (e.g., Cajochen et al., 2000; Phipps-Nelson et al., 2003), the current study showed that bright light can also counteract subjective fatigue during daytime and in the absence of sleep deprivation. In addition, the results on experiences, performance and physiology showed quite a complex picture, suggesting that the beneficial effects of exposure to bright light at night cannot be directly translated to daytime situations, but need further research. Moreover, our results highlight the challenge to optimize lighting scenarios supporting affective and cognitive functioning and providing pleasant conditions. As the findings showed that persons' sensitivity to alerting and vitalizing effects of bright light during daytime may depend on type of indicator, psychological state (i.e., fatigued vs. rested) and duration of exposure, this would motivate the development of dynamic and person-tailored lighting solutions to enhance alertness and vitality during the day.

In the current laboratory study, we compared two illuminance levels (i.e., 1000 lx vs. 200 lx at the eye). Exploring the daytime effects of a more varied range of intensities on subjective, performance and physiological markers of alertness and vitality will provide additional insights in persons' relative responsiveness to different intensity levels. In addition to intra-individual variations in a person's momentary state and type of activity, interpersonal differences in sensitivity to light effects (due for example to age, chronic fatigue or chronotype) should be considered when determining the optimal required light dosage throughout the day.

The light exposure in the current study was relatively brief (i.e., a half hour) and we did not measure alertness and vitality after the offset of the lighting conditions, so we cannot predict whether the effects would persist throughout the day. A few studies provided indications that light can affect subjective experiences of alertness as well as performance on a sustained attention task over longer periods of exposure (i.e., several hours) at night or during daytime after sleep or light deprivation (e.g., Lockley et al., 2006; Phipps-Nelson et al., 2003; Rüger et al., 2006). In addition, several studies suggested that light exposure during the day can influence alertness, performance and sleep later in the evening or at night (e.g., Figueiro, Nonaka, & Rea, 2013; Figueiro & Rea, 2010; Hubalek, Brink, & Schierz, 2010; Martinez-Nicolas, Ortiz-Tudela, Madrid, & Rol, 2011; Münch, Linhart, Borisuit, Jaeggi, & Scartezzini, 2012). Yet, whether exposure to bright light during the day results in, for example, prolonged vitality or instead results in exhaustion from over concentration is not known. Future research should thus investigate whether effects during the day will persist with long-term or repeated exposure as well as explore potential after-effects.

Although the current study provides new insights in potential alerting and vitalizing effects of bright light exposure during daytime, it also suggests that the impact of daytime light exposure on human behavior and experience requires more attention – as well as their potential underlying mechanisms – to establish optimal and pleasant lighting scenarios for daytime situations. Especially considering the potential environmental costs of energy expenditure, it is too early to start recommending the installation of more intense general lighting schemes to increase feelings of alertness and vitality. We do however recommend those in need of temporary mental restoration at work to seek exposure to daylight, either indoors near a window, or outdoors during a short break.

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